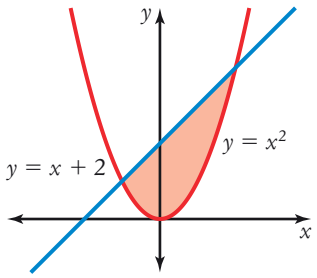
WORKSHEET

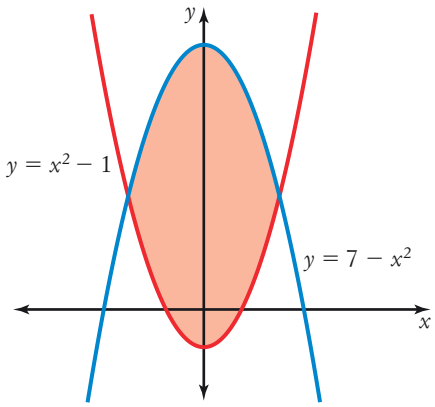
Chapter 6: Calculating areas between curves

1 Find the area enclosed between each pair of curves shown below.

a

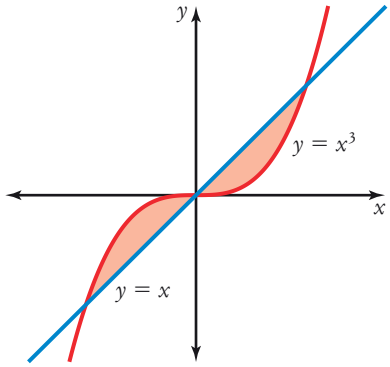


b

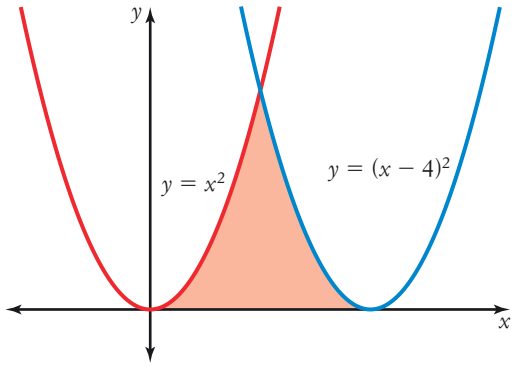


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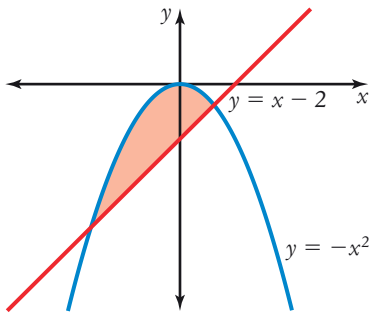
c



d

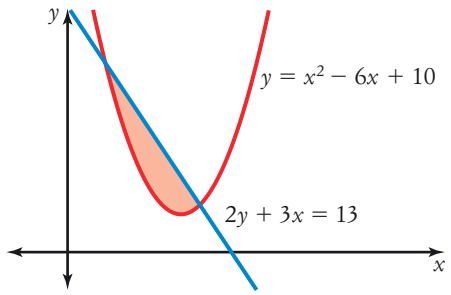


e



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f



2 Find the area bounded by the line $y = 1$ and the curve $y = x^2$.

3 Find the area bounded by the line $y = 2$ and the curve $y = x^2 + 1$.

4 Find the area bounded by the line $y = x$ and the curve $y = x^2$.

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5 Find the area bounded by the line $y = 5$ and the curve $y = 9 - x^2$.

6 Find the area bounded by the line $y = x + 6$ and the curve $y = x^2$.

7 Find the area bounded by the line $y = 4x$ and the curve $y = x^3$.

8 Find the area enclosed between the following pairs of functions.

a $y = (x - 1)^2$ and $y = (x + 1)^2$ and the x-axis.

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b $y = -6x + 16$ and $y = x^2$.

c $y = (x - 2)^2$ and $y = (x - 4)^2$ and the x-axis.

d $y = x^2$ and $y = x^3$.

9 CAS Use a graphics calculator to find the area bounded by the line $y = -3x + 4$, the x-axis and the curve $y = x^3$.

10 Find the exact area enclosed between the line $x - y + 4 = 0$ and the curve $y = \sqrt{4 - x^2}$.

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Answers

1 **a** $4\frac{1}{2}$ units^2

b $21\frac{1}{3}$ units^2

c $\frac{1}{2}$ units^2

d $5\frac{1}{3}$ units^2

e $4\frac{1}{2}$ units^2

f About 2.6 units^2

2 $1\frac{1}{3}$ units^2

3 $1\frac{1}{3}$ units^2

4 $\frac{1}{6}$ units^2

5 $10\frac{2}{3}$ units^2

6 $20\frac{5}{6}$ units^2

7 8 units^2

8 **a** $\frac{2}{3}$ units^2

b $166\frac{2}{3}$ units^2

c $\frac{2}{3}$ units^2

d $\frac{1}{12}$ units^2

9 0.42 units^2

10 $(\pi - 2)$ units^2

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